

The Berger–Coburn Conjecture: A Later Counterexample

Literature-status note for arXiv:2305.03948

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Status: literature already answered. The general conjecture is false; the quadratic-exponential special case in the source paper remains valid.

Original question signal

In the first paragraph of its introduction (PDF page 1), Xiong’s *Boundedness of metaplectic Toeplitz operators and Weyl symbols* [1] describes the Berger–Coburn conjecture as a long-standing open problem. In the standard Bargmann–Fock normalization, it asks whether a Toeplitz operator T_g with a measurable symbol and the natural kernel-domain condition extends boundedly if and only if the borderline heat transform $g^{(1/4)}$ —equivalently, the Weyl symbol—is bounded.

The source paper proves this equivalence when $g = e^q$ and q is a complex inhomogeneous quadratic polynomial subject to its stated nondegeneracy and integrability assumptions. Its abstract and introduction explicitly qualify the conclusion as completing the conjecture “in this case.”

Later answer

Looi’s *A counterexample to the Berger–Coburn conjecture* [2] restates the measurable-symbol assertion as Conjecture 1.1 on PDF pages 2–3 and then proves the following as Theorem 1.2 on PDF page 3:

For every complex dimension $n \geq 1$, there is a measurable symbol $g \in L^2(\mathbb{C}^n, d\mu)$ such that $gk_a \in L^2(d\mu)$ for every normalized kernel k_a , the Toeplitz form T_g extends to a bounded operator, but $g^{(1/4)}$ is unbounded.

This violates the necessity implication in the conjecture while satisfying its explicit hypotheses. The supporting paper also cites Xiong’s work among the metaplectic/quadratic settings in which the endpoint principle had been verified. The authors therefore explicitly know that they are disproving the general problem surrounding the source’s special-case theorem.

Scope and provenance

The later result is a full negative answer for general measurable symbols in every $n \geq 1$. It leaves Xiong’s positive characterization for quadratic-exponential symbols untouched.

The relation was found by searching the run indexes and current literature for the exact conjecture name, the source arXiv id, “Weyl symbol”, and the critical heat time. The supporting paper’s

title, Conjecture 1.1, Theorem 1.2, and citation to Xiong make this an explicit literature answer rather than an agent-created implication.

References

- [1] H. Xiong, *Boundedness of metaplectic Toeplitz operators and Weyl symbols*, arXiv:2305.03948; J. Funct. Anal. **286** (2024), 110294.
- [2] S. Looi, *A counterexample to the Berger–Coburn conjecture*, arXiv:2601.20859 (2026).